

Lesson

5

Cohesion and Coherence

It is a common Fault in Writers, to allow their Readers too much knowledge: They begin with that which should be the Middle, and skipping backwards and forwards, 'tis impossible for any one but he who is perfect in the Subject before, to understand their Work, and such an one has no Occasion to read it.

—BENJAMIN FRANKLIN

The two capital secrets in the art of prose composition are these: first, the philosophy of transition and connection; or the art by which one step in an evolution of thought is made to arise out of another: all fluent and effective composition depends on the connections; secondly, the way in which sentences are made to modify each other; for the most powerful effects in written eloquence arise out of this reverberation, as it were, from each other in a rapid succession of sentences.

—THOMAS DE QUINCEY

UNDERSTANDING HOW SENTENCES CONNECT

So far, I have focused mainly on individual sentences, and I have treated clarity as if we could achieve it just by mapping characters and actions onto subjects and verbs. But readers need more than individually clear sentences before they feel a passage “hangs together.” These two passages, for example, say much the same thing but feel very different:

1a. The safeguard of democracy everywhere—an educated citizenry—is being threatened by college costs that have been rising fast for the last several years. Increases in family income have been significantly outpaced by increases in tuition at colleges and universities during that period. Only the children of the wealthiest families in our society will be able to afford a college education if this trend continues. Knowledge and intellectual skills, in addition to wealth, will divide us as a people, when that happens. Equal opportunity and the egalitarian basis of our democratic society could be eroded by such a divide.

✓ 1b. In the last several years, college costs have been rising so fast that they are now threatening the safeguard of democracy everywhere: an educated citizenry. During that period, tuition has significantly outpaced increases in family income. If this trend continues, a college education will soon be affordable only by the children of the wealthiest families in our society. When that happens, we will be divided as a people not only by wealth, but by knowledge and intellectual skills. Such a divide will erode equal opportunity and the egalitarian basis of our democratic society.

The first seems choppy, even disorganized; the second seems more connected.

But like the word *clear*, the words *choppy*, *disorganized*, and *connected* refer not to the words on the page but to how they make us feel. What is it about the *arrangement* of words in (1a) that makes us feel we are moving through it in fits and starts? Why does (1b) seem to flow more easily? We base those judgments on two aspects of word order:

- We judge a sequence of sentences to be *cohesive* based on how each sentence ends and the next begins.
- We judge a whole passage to be *coherent* based on how all the sentences in it cumulatively begin.

In this lesson, I discuss the cohesion and coherence of passages; in Lessons 7 and 8, I discuss the coherence of whole documents.

COHESION

The Sense of Flow

In Lesson 4, we devoted a few pages to that familiar advice, *Avoid passives*. If we always did, we would choose the active verb in sentence (2a) over the passive in (2b):

2a. The collapse of a dead star into a point perhaps no larger than a marble **CREATES**_{active} a black hole.

2b. A black hole **IS CREATED**_{passive} by the collapse of a dead star into a point perhaps no larger than a marble.

But we might choose otherwise in context. Consider:

¹Some astonishing questions about the nature of the universe have been raised by scientists studying black holes in space. ^{2a}**The collapse of a dead star into a point perhaps no larger than a marble creates a black hole.** ³So much matter compressed into so little volume changes the fabric of space around it in puzzling ways.

¹Some astonishing questions about the nature of the universe have been raised by scientists studying black holes in space. ^{2b}**A black hole is created by the collapse of a dead star into a point perhaps no larger than a marble.** ³So much matter compressed into so little volume changes the fabric of space around it in puzzling ways.

In this context, our sense of “flow” calls not for (2a), the sentence with the active verb, but for (2b), the one with the passive.

The reason is clear: the last four words of sentence (1) introduce an important character—*black holes in space*. But with sentence (2a), the next concepts we hit are *collapsed stars* and *marbles*, information that seems to come out of nowhere:

¹Some astonishing questions about the nature of the universe have been raised by scientists studying black holes in space. ^{2a}**The collapse of a dead star into a point perhaps no larger than a marble creates.** . . .

If, however, we follow sentence (1) with (2b), the sentence with the passive verb, we feel those sentences connect more smoothly, because now the first words in (2b) repeat what we just read at the end of (1):

¹. . . by scientists studying black holes in space. ^{2b}**A black hole** is created by . . .

Note too that the passive lets us put at the *end* of sentence (2b) words that connect it to the *beginning* of sentence (3):

¹. . . black holes in space. ^{2b}**A black hole** is created by the collapse of a dead star into **a point perhaps no larger than a marble**.

³**So much matter compressed into so little volume** changes the fabric of space around it in puzzling ways.

Here's the point: Sentences are *cohesive* when the last few words of one sentence set up information that appears in the first few words of the next. That's what gives us our experience of flow. And in fact, that's the main function of the passive in the language: to let us arrange sentences so that they flow easily from one to the next.

Managing Information: Old Before New

We learn by connecting new information to what we already know. In sentences, therefore, readers prefer to encounter information that is old or familiar *to them* before they encounter information that is new or unfamiliar. So:

1. **Begin sentences with information familiar to your readers.**

Readers get that familiar information in two ways. First, they remember words from the sentences they just read. That's why in our example about black holes, the beginning of (2b) coheres with the end of (1) and why the beginning of (3) coheres with the end of (2b). Second, readers bring to a sentence a general knowledge of its subject. We would not be surprised, for example, to find the next sentence (4) begin like this:

. . . changes the fabric of space around it in puzzling ways.
⁴**Astronomers have reported** that . . .

The word *Astronomers* did not appear in the preceding sentences, but since we are reading about space, we wouldn't be surprised to find it beginning a sentence in the passage.

2. **End sentences with information that readers cannot anticipate.** Your sentences have to tell your readers something new, but readers always prefer to receive this new information after they have read something familiar.3. **Begin sentences with information that readers will find simple; end with information they will find complex.** This guideline follows from the others. What's familiar to your readers will seem simple to them; what's unfamiliar will seem complex.

You can more easily see when others fail to observe this old-before-new principle than when you do, because after you've

worked on your own ideas for a while, they all seem familiar—to you. But hard as it is to distinguish old from new in your own writing, you have to try, because readers expect sentences to begin with information that is familiar to *them* and to end with information that is new. Thwart this expectation too often and your readers won't understand you (or at least think they don't).

In every sequence of sentences you write, you have to balance principles that make individual sentences clear and principles that make the whole passage cohesive. *But in that tradeoff, give priority to helping readers create a sense of cohesive flow.* Fortunately, the principle of old before new cooperates with the principle of characters as subjects. Once you mention your main characters, readers recognize them as familiar. So when characters are up front, so is familiar information.

Here's the point: So far, we have identified three main principles of clarity. Two are about sentences:

- Make main characters the subjects of sentences.
- Make important actions verbs.

The third is about sentences as well, but it also explains how sentences flow together:

- Put old information before new information.

These principles usually complement one another, but if you have to choose among them, favor the third. The way you organize old and new information determines how cohesive readers will find your writing. And for readers, a passage's overall *cohesion* trumps the *clarity* of individual sentences.

QUICK TIP Writers often refer to something in a previous sentence with words such as *this*, *these*, *that*, *those*, *another*, *such*, *second*, or *more*. When you use any of those signals, try to put them at or close to the beginning of a sentence:

How to calculate credits for classes taken in a community college is **another** issue that we must consider.

- ✓ **Another** issue that we must consider is how to calculate credits for classes taken in a community college.

Exercise 5.1

Revise these two passages to improve their flow by putting old information first in each sentence. In (1), I have boldfaced the words that I feel are old information.

1. Two aims—the recovery of the American economy and the modernization of America into a military power—were in the **president's mind when he assumed office**. The drop in unemployment figures and inflation, and the increase in the GDP testifies to **his success in the first**. But America's increased involvement in international conflict without any clear set of political goals indicates **less success with the second**. Nevertheless, increases in the military budget and a good deal of saber rattling **pleased the American voter**.
2. The components of Abco's profitability, particularly growth in Asian markets, will be highlighted in our report to demonstrate its advantages versus competitors. Revenue returns along several dimensions—product type, end-use, distribution channels, etc.—will provide a basis for this analysis. Likely growth prospects of Abco's newest product lines will depend most on the development of distribution channels in China, according to our projections. A range of innovative strategies will be needed to support the introduction of new products.

COHERENCE

A Sense of the Whole

When you create cohesive flow, you take the first step toward helping readers feel that your prose hangs together. But they will judge you to be a competent writer only when they also feel that your writing has *coherence*, a quality different from *cohesion*. It's easy to confuse these words because they sound alike.

- *Cohesion* is when pairs of sentences fit together the way two pieces of a jigsaw puzzle do (recall the black hole example).
- *Coherence* is when all the sentences in a piece of writing add up to a larger whole, the way all the pieces in a puzzle add up to the picture on the box.

This next passage has good cohesive flow because we move from one sentence to the next without a hitch:

Sayner, Wisconsin, is the snowmobile capital of the world. The buzzing of snowmobile engines fills the air, and their tank-like tracks crisscross the snow. The snow reminds me of Mom's mashed

potatoes, covered with furrows I would draw with my fork. Her mashed potatoes usually make me sick—that's why I play with them. I like to make a hole in the middle of the potatoes and fill it with melted butter. This behavior has been the subject of long chats between my analyst and me.

Though its individual sentences are cohesive, that passage as a whole is incoherent. (It was created by six different writers, one of whom wrote the first sentence, with the other five sequentially adding one sentence, knowing only the immediately preceding one.) It is incoherent for three reasons:

1. The subjects of the sentences are entirely unrelated.
2. The sentences share no common themes or ideas.
3. The paragraph has no one sentence that states what the whole passage is about.

I will discuss that second point in Lesson 6 and the third one in Lesson 8. The rest of this lesson focuses on the first point, shared subjects.

Subjects and Topics

For 500 years, English teachers have defined *subject* in two ways:

1. The "doer" of the action
2. What a sentence is "about" or "comments" on, its TOPIC

In Lessons 3 and 4, we saw why that first definition doesn't work: the subjects of many sentences are not doers. Here, for example, the subject is an action: *The explosion was loud*. Here it is a quality: *Correctness is not writing's highest virtue*. Here it is just a grammatical placeholder: *It was a dark and stormy night*.

But also flawed is that second definition: *A subject is what a sentence is about*. It is flawed because, often, a sentence's topic is stated elsewhere than in the grammatical subject.

For example, none of the MAIN SUBJECTS in these sentences names their topics.

- The main subject of this sentence is *it*, but its topic is *your claim*, the object of the preposition *for*:

It is impossible for your claim to be proved.

- The subject of this sentence is *I*, but its topic is *this question*, the object of *to*:

In regard to this question, I believe more research is needed.

- The subject of this sentence is *it*, but its topic is *our proposal*, the subject of a verb in a subordinate clause:

It is likely that our proposal will be accepted.

- The subject of this sentence is *no one*, but its topic is *such results*, a direct object shifted to the front for emphasis:

Such results no one could have predicted.

Topics and Coherence

A sentence's topic does not have to be its grammatical subject, although in writing that is clear and coherent, it often is. *Topic* is not a grammatical term but a psychological one: it refers to the idea that readers expect the sentence to be "about" or "comment" on. Readers expect to find this idea stated toward the beginning of the sentence.

Readers consider a passage coherent to the degree that they can quickly and easily see two things:

- the topics of individual sentences and clauses
- how the topics in the passage make up a related set of concepts

How does this passage strike you?

Consistent ideas toward the beginnings of sentences, especially in their subjects, help readers understand what a passage is generally about. A sense of coherence arises when a sequence of topics comprises a narrow set of related ideas. But the context of each sentence is lost by seemingly random shifts of topics. Unfocused paragraphs result when that happens.

The passage seems choppy and disorganized because the topics of its sentences are inconsistent and diffuse. They do not focus our attention on a limited set of related ideas:

Consistent ideas toward the beginnings of sentences, especially in their subjects, help readers understand what a passage is generally about. A sense of coherence arises when a sequence of topics comprises a narrow set of related ideas. But the context of each sentence is lost by seemingly random shifts of topics. Unfocused, even disorganized paragraphs result when that happens.

Now compare that passage to this revision, with the new topics boldfaced:

Readers understand what a passage is generally about when **they** see consistent ideas toward the beginnings of sentences, especially

in their subjects. **They** feel a passage is coherent when **they** read a sequence of topics that focuses on a narrow set of related ideas. But when **topics** seem to shift randomly, **readers** lose the context of each sentence. When **that** happens, **they** feel they are reading paragraphs that are unfocused and even disorganized.

The subjects of sentences and clauses focus our attention on just two concepts—*readers* and *topics*—and form a strong TOPIC STRING: *readers, they, they, they, topics, readers, that, they [readers]*. That is why this passage seems more coherent.

How to Revise: Topics

Here is how to analyze and revise your writing so it is coherent.

1. Analyze

- Underline the first seven or eight words of every sentence in a passage, stopping when you hit a verb.
- If you can, underline the first five or six words of every clause in those sentences.

2. Assess

- Do the underlined words constitute a relatively small set of related ideas? Even if you see how they are related, will your readers? If you answer *no*, you should revise.
- Do the underlined words name the most important characters, real or abstract? Again, if you answer *no*, you should revise.
- Imagine giving the passage a title. The words in it are likely to name important topics.

3. Rewrite

- In most of your sentences, although not necessarily all, use subjects to name topics.
- Put these subjects/topics close to the beginnings of your sentences.
- Be sure that those topics are, in context, familiar to your readers.

QUICK TIP When you start to draft a new section of your paper, list the topics and characters you intend to write about. Include not just flesh-and-blood characters, but

important concepts as well. As you draft, try to put those characters into the subjects of most of your sentences. If you do not mention one of those characters for several sentences, you may have gotten off track.

Likewise, when you must read a passage you expect to be difficult, first skim it quickly to find its main topics and characters. Think about them for a moment. The more sharply you have them in mind, the more easily you will understand the passage.

BEGINNING SENTENCES WELL

It is hard to begin a sentence well. Readers want to get to a subject/topic quickly, but too often we begin sentences in ways that keep readers from doing that. It's called *throat-clearing*. Throat-clearing typically begins with metadiscourse (review pp. 58–59) that connects a sentence to the previous one, with transitions such as *and*, *but*, *therefore*:

And therefore . . .

We then add a second kind of metadiscourse that expresses our attitude toward what is coming, words such as *fortunately*, *perhaps*, *allegedly*, *it is important to note*, *for the most part*, or *in a manner of speaking*:

And therefore, it is important to note . . .

Then we indicate time, place, or manner:

And therefore, it is important to note that, in Eastern states since 1980 . . .

Only then do we get to the subject/topic:

And therefore, it is important to note that, in Eastern states since 1980, **acid rain** has become a serious problem.

When you open several sentences like that, your readers have a hard time seeing not just what each sentence is about, but the focus of the whole passage. When you find a sentence with lots of words before its subject/topic, revise:

- ✓ Since 1980, therefore, **acid rain** has become a serious problem in the Eastern states.

Here's the point: Before you begin writing, name the things you are writing about. Those are your *topics*. They should be short, concrete, familiar words, and more often than not, they should name the main characters in your story. Most of your subjects should be topics. And be consistent: do not vary your subjects for the sake of variety. Together, your subjects should name the topics that tell your readers what a passage as a whole is "about."

Exercise 5.2

Revise these passages to give them more consistent topic strings. Identify words that name what the passages "comment" on and use those words as subjects of most of the sentences. In (1), words that could be consistent subjects/topics are boldfaced.

1. **Vegetation** covers the earth, except for those areas continuously covered with ice or utterly scorched by continual heat. Richly fertilized plains and river valleys are places where **plants** grow most richly, but also at the edge of perpetual snow in high mountains. The ocean and its edges as well as in and around lakes and swamps are **densely vegetated**. The cracks of busy city sidewalks have **plants** in them as well as in seemingly barren cliffs. Before humans existed, the earth was covered with **vegetation**, and the earth will have **vegetation** long after evolutionary history swallows us up.
2. The power to create and communicate a new message to fit a new experience is not a competence animals have in their natural states. Their genetic code limits the number and kind of messages that they can communicate. Information about distance, direction, source, and richness of pollen in flowers constitutes the only information that can be communicated by bees, for example. A limited repertoire of messages delivered in the same way, for generation after generation, is characteristic of animals of the same species, in all significant respects.

TWO QUALIFICATIONS

Alleged Monotony

A common piece of advice is, *Vary how you begin your sentences*. That's a bad idea, especially when you change subjects just to make them different. When you see the same topic in several sentences in your own prose you might think a passage is monotonous. But

your readers may be grateful for the repetition, because it will help them focus on your ideas.

On the other hand, you might revise if you find you have used exactly the same words for the same topics in exactly the same positions. This passage goes over the top in that kind of consistency:

"**Moral climate**" is created when an objectivized moral standard for treating people is accepted by others. **Moral climate** results from norms of behavior that are accepted by society whereby if people conform they are socially approved of, or if they don't they are shunned. In this light, **moral climate** acts as a reason to refrain from saying or doing things that the community does not support. **A moral climate** encourages individuals to conform to a moral standard and apply that standard to their own circumstances.

In such passages, you can vary a few of the words that refer to a repeated topic:

"**Moral climate**" is created. . . . **This climate** results. . . . In this light, **morality** acts. . . . **A moral climate** encourages. . . .

Be cautious, though: most writers change topics too often.

Faked Coherence

Some writers try to fake coherence by lacing their prose with conjunctions like *thus*, *therefore*, *however*, and so on, regardless of whether they signal real logical connections. An example:

Because the press is the major medium of interaction between the president and the people, how it portrays him influences his popularity. **Therefore**, it should report on the president objectively. Both reporters and the president are human, **however**, subject to error and favoritism. **Also**, people act differently in public than they do in private. **Hence**, to understand a person, it is important to know the whole person, his environment, upbringing, and education. **Indeed**, from the correspondence with his family, we can learn much about Harry S. Truman, our thirty-third president.

Experienced writers use these connecting devices, but they depend more on the logical flow of their ideas. They are especially careful not to overuse words like *and*, *also*, *moreover*, *another*, and so on, words that say simply *Here's one more thing*. You need a *but* or *however* when you contradict or qualify what you just said, and you can use a *therefore* or *consequently* to wind up a line of reasoning. But avoid using words like these more than a few times a page. Your readers don't need them when your sentences are cohesive and the passage they make up is coherent.

IN YOUR OWN WORDS

Exercise 5.3

Writers often violate the principle that old information should come before new information because they know their own writing too well: to them, everything can seem like old information. So work with a reader to analyze the flow of old and new information in a passage of your writing. Have a reader go through a passage of your writing and underline every piece of new information. If the beginnings of sentences are underlined, you need to revise. Do so.

Exercise 5.4

In Lesson 4 (p. 55), I noted that you could change the point of view of a passage by changing the characters/topics that appear as subjects in its sentences:

By early 1945, *the Allies* had essentially defeated *Germany* . . .

By early 1945, *Germany* had essentially been defeated by *the Allies* . . .

The first version is written from the point of view of the Allies, the second from that of Germany.

Experiment with the point of view of a passage of your own writing. First, circle words that name characters/topics, wherever they appear. Then underline the subject of every clause. You should see that some characters/topics appear most often as subjects, while others appear most often in other parts of your sentences (likely after the verb). Revise the passage by using those other characters/topics as subjects and by moving characters/topics used as subjects after the verb. What changes do you notice in the feel or even meaning of the passage?

Exercise 5.5

Writers use conjunctions and transitions like *also*, *furthermore*, *moreover*, *another*, *but*, *however*, *although*, *nevertheless*, and *consequently* to help readers see the connections among their ideas. But such words can also be used to bluff, to fake connections that aren't really there. Writers are most likely to fake connections when they are struggling to figure out or to express their ideas. Select a piece of writing you are struggling with, and have a

reader call your bluffs. Ask your reader to circle words that assert logical connections that don't seem to be there. Then revise as necessary. Work to refine not only your writing but also your ideas and thinking.

SUMMING UP

I noted in Lessons 3 and 4 that to understand a sentence, readers have to integrate two levels of sentence structure: the grammar level and the story level. I now add a third level, the information level. Writers can put familiar information and unfamiliar information anywhere in a sentence, but readers prefer to find these kinds of information in particular places. Just as readers prefer characters to align with subjects and actions with verbs, so they expect to find old, familiar information toward the beginning of a sentence, aligned with its topic, and new, unfamiliar information toward the end. We can graphically integrate these principles with our others. (I'll fill in the empty box in Lesson 6):

Fixed Positions	Topic			
Movable Elements	Old/Familiar		New/Unfamiliar	Information Level
Fixed Positions	Subject	Verb		Grammar Level
Movable Elements	Character	Action		Story Level

The principles from this lesson are these:

1. Begin sentences with subjects or, sometimes, short introductory phrases that communicate old information, information that your readers are familiar with (boldfaced); give new, unfamiliar information (italicized) toward the ends of sentences:

The number of dead in the Civil War exceeded **all other wars in American history** combined. A reason for *the lingering animosity between North and South* today is **the memory of this terrible carnage**.

- ✓ Of **all the wars in American history**, none has exceeded the Civil War in *the number of dead*. **The memory of this terrible carnage** is one reason for *the lingering animosity between North and South* today.

2. Through a series of sentences, keep your topics short and reasonably consistent:

Competition between Asian and American companies in the Pacific is the first phase of this study. **Labor costs and the ability to introduce new products quickly in particular** are examined. **A plan that will show American industry how to restructure its facilities** will be developed from this study.

- ✓ In the first phase of this study, **we** examine how **Asian and American companies** compete in the Pacific region. **We** examine in particular their labor costs and ability to introduce new products quickly. **We** develop from this study a **plan** that will show **American industry** how to restructure its facilities.

Lesson

6

Emphasis

In my end is my beginning.

—T. S. ELIOT

All's well that ends well.

—WILLIAM SHAKESPEARE

UNDERSTANDING HOW SENTENCES END

If you consistently write sentences whose subjects name a few central characters and topics, and you join them to strong verbs, you'll likely get the rest of the sentence right and, in the process, create a passage that is both cohesive and coherent. But if the first few words of a sentence are worth special attention, so are the last few. How you end your sentences affects how readers judge not only the clarity and strength of individual sentences, but also their collective cohesion and coherence.

When readers build up momentum in the first nine or ten words of a sentence, they more easily get through complicated material that follows. Compare:

1a. A sociometric and actuarial analysis of Social Security revenues and disbursements for the last six decades to determine changes in projecting deficits is the subject of this study.

- ✓ 1b. In this study, we analyze Social Security's revenues and disbursements for the last six decades, using sociometric and actuarial criteria to determine changes in projecting deficits.

As we start (1a), we struggle to understand its technical terms at the same time we are hacking through a subject twenty-two words long. In (1b), we go through just five words to get past a subject and verb and eleven more before we hit a term—sociometric—that might slow us down. By that point we have enough momentum to carry us through the complexity to the sentence's end.

Complex Grammar

Which of these two sentences do you prefer?

2a. Lincoln's claim that the Civil War was God's punishment of both North and South for slavery appears in the last part of the speech.

2b. In the last part of his speech, Lincoln claims that God gave the Civil War to both North and South as a punishment for slavery.

Most readers prefer (2b), because it begins simply with a short introductory phrase followed by a one-word subject and a specific verb, and then moves toward grammatical complexity. We discussed that issue in Lesson 5.

Complex Meaning

Another kind of complexity is in the meanings of words, especially technical terms. Compare these two passages:

3a. The role of calcium blockers in the control of cardiac irregularity can be seen through an understanding of the role of calcium in the activation of muscle cells. The regulatory proteins actin, myosin, tropomyosin, and troponin make up the sarcomere, the basic unit of muscle contraction. The energy-producing, or ATPase, protein myosin makes up its thick filament, while actin, tropomyosin, and troponin make up its thin filament. Interaction of myosin and actin triggers muscle contraction.

- ✓ 3b. When a muscle contracts, it uses calcium. We must therefore understand how calcium affects muscle cells to understand how cardiac irregularity is controlled by drugs called calcium blockers. The basic unit of muscle contraction is the sarcomere. It has two filaments, one thin and one thick. Those filaments consist of four proteins that regulate contraction: actin, tropomyosin, and troponin in the thin filament and myosin in the thick one. Muscles contract when the protein actin in the thin filament interacts with myosin, an energy-producing or ATPase protein in the thick filament.

Both passages use the same technical terms, but (3b) is clearer to those who know nothing about the chemistry of muscles.

Those passages differ in two ways. First, information that is only implicit in (3a) is stated explicitly in (3b). More important, almost all the technical terms in (3a) are toward the beginnings of sentences and clauses, increasing the passage's feeling of complexity:

3a. The role of **calcium blockers** in the control of **cardiac irregularity** can be seen through an understanding of the role of calcium in the activation of muscle cells.

The **regulatory proteins actin, myosin, tropomyosin, and troponin** make up the **sarcomere**, the basic unit of muscle contraction.

The **energy-producing, or ATPase, protein myosin** makes up its thick filament, while **actin, tropomyosin, and troponin** make up its thin filament.

Interaction of myosin and actin triggers muscle contraction.

In (3b), those technical terms appear toward the ends of sentences:

... uses **calcium**.

... is controlled by drugs called **calcium blockers**.

... is the **sarcomere**.

... four proteins that regulate contraction: **actin, tropomyosin, and troponin** in the thin filament and **myosin** in the thick one.

... **myosin**, an **energy-producing or ATPase protein** in the thick filament.

This way of introducing unfamiliar terms works even for prose intended for professional readers. In this next passage, from the *New England Journal of Medicine*, the writer deliberately uses metadiscourse just to put the new technical term at the end:

The incubation of peripheral-blood lymphocytes with a lymphokine, interleukin-2, generates lymphoid cells that can lyse fresh, noncultured, natural-killer-cell-resistant tumor cells but not normal cells. *We term these cells lymphokine-activated killer (LAK) cells.*

Here's the point: Your readers want you to organize your sentences to help them manage two kinds of difficulty:

- long and complex phrases and clauses
- new information, particularly unfamiliar technical terms

In general, your sentences should begin with elements that are relatively short: a short introductory phrase or clause, followed by a short, concrete subject, followed by a verb expressing a specific action. After the verb, the sentence can go on for several lines, if it is well constructed (see Lessons 10 and 11). The general principle is to carry the reader not from complexity to simplicity but from simplicity to complexity.

EMPHASIS AND STRESS

In the last lesson, we said that the first few words of a sentence are important because they state its *topic*, what the sentence is “about” or “comments” on. The last few words of a sentence are also important, because they receive special emphasis. You can sense that when you hear your voice rise at the end of a sentence to emphasize one syllable more strongly than the others:

... more strongly than the 6-thers.

We’ll call this most emphatic part of a sentence its **STRESS**.

How you manage that stress position helps establish the voice readers hear in your prose, because if you end a sentence on words that carry little meaning, your sentence will seem to end weakly:

Climate change could raise sea levels to a point where much of the world’s low-lying coastal areas would disappear, **according to most atmospheric scientists.**

- ✓ According to most atmospheric scientists, climate change could raise sea levels to a point where much of the world’s low-lying coastal areas **would disappear.**

In Lessons 4 and 5, we saw how different subjects/topics create different points of view (pp. 55–56, 71–73). You can also manage your endings to emphasize important themes.

Compare these passages. The first laments the way universities in the 1990s responded to a decline in the number of potential students—by becoming increasingly commercial:

The universities were going to have to pursue students much **as businesses pursue customers.** They were going to have to treat their prospective students **as potential buyers.** And they were going to have to treat their existing students **as customers too**, for students **can always switch brands.**

—Mark Edmundson, *Why Teach? In Defense of a Real Education*

Moving those references to businesses, customers, buyers, and brands out of the sentences’ ends blunts the passage’s edge:

The universities were going to have to act like **businesses pursuing customers** to attract prospective students. And because **customers can always switch brands**, universities were going to have to take the same approach with their existing students too.

In some cases, changing a sentence’s stress can even change its meaning. Which company would you invest in?

Although the company’s sales remain strong, **its stock price has slipped.**

Although the company’s stock price has slipped, **its sales remain strong.**

According to that first sentence, the company is in trouble; according to the second, it’s a bargain.

Here’s the point: Just as we look to the first few words of a sentence for point of view, so we look to the last few words for special emphasis. You can revise a sentence to emphasize particular words that you want readers to hear stressed and thereby note as particularly significant.

HOW TO REVISE: STRESS

If you have managed your subjects and topics well, you will by default put the words you want to emphasize toward the ends of your sentences. To test this, read your sentence aloud and, as you reach the last three or four words, tap your finger hard as if emphasizing them in a speech. If you tap on words that do not deserve strong emphasis, look for words that do. Then put those words closer to the end. Here are some ways to do that.

Three Tactical Revisions

1. Trim the end.

Sociobiologists claim that our genes control our social behavior **in the way we act in situations we are in every day.**

Since *social behavior* means *the way we act in situations . . .*, we can drop everything after *behavior*:

- ✓ Sociobiologists claim that our genes **control our social behavior.**

2. Shift peripheral ideas to the left.

The data offered to prove ESP are weak, **for the most part**.

- ✓ **For the most part**, the data offered to prove ESP are **weak**.

Particularly avoid ending with anticlimactic metadiscourse:

Job opportunities in computer programming are getting scarcer, **it must be remembered**.

- ✓ **It must be remembered** that job opportunities in computer programming are getting scarcer.

3. Shift new information to the right. A more common way to manage stress is by moving new information to the end of a sentence:

Questions about the ethics of withdrawing intravenous feeding are more difficult [than something just mentioned].

- ✓ More difficult [than something just mentioned] are **questions about the ethics of withdrawing intravenous feeding**.

Six Syntactic Devices to Emphasize the Right Words

There are a number of syntactic devices that let you manage where in a sentence you stress units of new information. (You just read one of them.)

1. **There shift** Some editors discourage all *there is/there are* constructions, but using them lets you shift a subject to the right to emphasize it. Compare:

Several syntactic devices let you manage where in a sentence you locate units of new information.

- ✓ *There are* **several syntactic devices** that let you manage where in a sentence you locate units of new information.

Experienced writers commonly begin a paragraph with *there* to introduce new topics and concepts that they develop in sentences that follow.

2. **Passives (for the last time)** A passive verb lets you flip a subject and object to get old and new information in the right order. Compare these sentences:

Some claim that **our genes** influence ^{active} aspects of behavior that we think are learned. **Our genes**, for example, seem to determine. . .

- ✓ Some claim that aspects of behavior that we think are learned are in fact influenced ^{passive} **by our genes**. **Our genes**, for example, seem to determine. . .

3. **What shift** This is another device that shifts a part of the sentence to the right, thereby emphasizing it more:

We need a monetary policy that would end fluctuations in money supply, unemployment, and inflation.

- ✓ **What we need is** a monetary policy that would end fluctuations in money supply, unemployment, and inflation.

4. **It shift** When you have a subject consisting of a long NOUN CLAUSE, you can move it to the end of the sentence and start with an *it*:

That oil prices would be set by OPEC once seemed inevitable.

- ✓ *It* once seemed inevitable **that oil prices would be set by OPEC**.

5. **Not only X, but (also) Y (as well)** In this next pair, note how the *but* emphasizes the last element of the pair:

We must clarify these issues and **develop trust**.

- ✓ We must *not only* clarify these issues *but also* **develop trust**.

Unless you have a reason to emphasize the negative, end with the positive:

The point is to highlight our success, **not to emphasize our failures**.

- ✓ The point is not to emphasize our failures but **to highlight our success**.

6. **Pronoun substitution and ellipsis** This is a fine point: a sentence can end flatly when you repeat a word that you used just a few words before at the end of a sentence, because the voice we hear in our mind's ear drops off at the end. If you read aloud the preceding sentence, this one, and the next, you can hear that drop at the end of each sentence. To avoid that kind of flatness, rewrite or use a pronoun instead of repeating the word at the end of the sentence. For example:

A sentence will seem to end flatly if at its end you use a word that you used just a few words before, because when you repeat that word, your voice **drops**. Instead of repeating the noun, use a **pronoun**. The reader will at least hear emphasis on the word just **before it**.

Occasionally, you can just delete words that repeat earlier ones:

It is sometimes possible to represent a complex idea in a simple sentence, but more often you cannot.

One of the characteristics of especially elegant prose is how writers use a handful of devices to end their sentences. I will discuss those in Lesson 11.

QUICK TIP You can easily check whether you have stressed the right words by reading your sentences aloud: as you speak the last few words, raise your voice and tap the table with your fingers. If you've stressed the wrong words, your voice and table thumping will feel wrong:

It is sometimes possible to represent a complex idea in a simple sentence, but more often you cannot represent it in that kind of sentence.

If you've stressed the right words, your voice and table thumping will feel right:

It is sometimes possible to represent a complex idea in a simple sentence, but **MORE OFTEN YOU CANNOT.**

Exercise 6.1

Revise these sentences to emphasize the right words. In the first three, I boldfaced what I think should be stressed. Then eliminate wordiness, nominalizations, etc.

1. The President's tendency to **rewrite the Constitution** is the biggest danger to America, in my opinion, at least.
2. A new political philosophy that could affect our society **well into the twenty-first century** may emerge from these studies.
3. There are **limited** opportunities for faculty to work with individual students in large colleges and universities.
4. Building suburban housing developments in floodplains has led to the existence of extensive and widespread flooding and economic disaster in parts of our country in recent years, it is now clear.
5. The teacher who makes an assignment of a long final term paper at the end of the semester and who then gives only a grade and nothing else such as a critical comment is a common object of complaint among students at the college level.
6. Renting textbooks rather than buying them for basic required courses such as mathematics, foreign languages, and English, whose textbooks do not go through yearly changes, is feasible, however, economically speaking.

Exercise 6.2

Revise these passages so that their sentences begin with appropriate topics and end with appropriate emphasis.

1. Athens's catastrophic Sicilian Invasion is the most important event in Thucydides's *History of the Peloponnesian War*. Three-quarters of the history is devoted to setting up the invasion because of this. Through the step-by-step decline in Athenian society that Thucydides describes, we can see how he chose to anticipate the Sicilian Invasion. The inevitability that we associate with the tragic drama is the basic reason for the need to anticipate the invasion.
2. Whether the date an operation intends to close down might be part of management's "duty to disclose" during contract bargaining is the issue here, it would appear. The minimization of conflict is the central rationale for the duty that management has to bargain in good faith. In order to allow the union to put forth proposals on behalf of its members, companies are obligated to disclose major changes in an operation during bargaining, though the case law is scanty on this matter.

TOPICS, STRESS, THEMES, AND COHERENCE

There is one more function performed by the stress of certain sentences, one that helps readers think a whole passage is coherent. As we saw in the last lesson, readers take the clearest topic to be a short noun phrase that comes early in a sentence, usually as its subject. That's why most of us judge this next paragraph to be unfocused: its sentences seem to open randomly, from no consistent point of view:

- 1a. Great strides in the early and accurate diagnosis of Alzheimer's disease have been made in recent years. Not too long ago, senility in an older patient who seemed to be losing touch with reality was often confused with Alzheimer's. Genetic clues have become the basis of newer and more reliable tests in the last few years, however. The risk of human tragedy of another kind, though, has resulted from the increasing accuracy of these tests: predictions about susceptibility to Alzheimer's have become possible long before the appearance of any overt symptoms. At that point, an apparently healthy person could be devastated by such an early diagnosis.

If we revise that paragraph to make the topics more consistent, we also make it more coherent (topics are boldfaced):

- ✓ 1b. In recent years, **researchers** have made great strides in the early and accurate diagnosis of Alzheimer's disease. Not too long ago, when

a **physician** examined an older patient who seemed out of touch with reality, **she** had to guess whether the **person** was senile or had Alzheimer's. In the past few years, however, **physicians** have been able to use new and more reliable tests focusing on genetic clues. But in the **accuracy of these new tests** lies the risk of another kind of human tragedy: **physicians** may be able to predict Alzheimer's long before its overt appearance, but **such an early diagnosis** could psychologically devastate an apparently healthy person.

The paragraph now focuses on just two topics: researchers/physicians and testing/diagnosis.

But there is one more revision that would make it more coherent still. Readers expect the opening sentence or two of a passage to announce its key concepts, and they look for those concepts in the last few words of those opening sentences, especially the first. Therefore, follow this principle:

Put key words in the stress position of a passage's *first* sentence to emphasize the key concepts it will repeat and develop.

The first sentence of that revised paragraph stresses advances in diagnosis: . . . *the early and accurate diagnosis of Alzheimer's disease*. The passage, however, is not about diagnosis but its risks. That organizing concept, though, does not appear until more than halfway through the paragraph. Readers would grasp the point of the paragraph better if all of its key concepts appeared in the first sentence, *specifically toward its end, in its stress position*.

Here is a new first sentence for the paragraph that would help readers focus on the key concepts not just of *Alzheimer's* and *new diagnoses*, but of *new problem* and *informing those most at risk*:

In recent years, researchers have made great strides in the early and accurate diagnosis of Alzheimer's disease, but those **diagnoses** have raised a **new problem** about **informing those most at risk** who **show no symptoms** of it.

We can call those key concepts that run through a passage its **THEMES**.

Look at the highlighted words in the passage below one more time:

- The boldfaced words are about testing.
- The italicized words are about mental states.
- The capitalized words are about a new problem.

Each of those concepts is announced toward the end of the new opening sentence, especially the theme of the new problem.

✓ 1c. In recent years, researchers have made great strides in the early and accurate **diagnosis** of *Alzheimer's disease*, but those **diagnoses** have raised A NEW PROBLEM about INFORMING THOSE MOST AT RISK WHO SHOW NO SYMPTOMS OF IT. Not too long ago, when a physician examined an older patient who seemed out of touch with reality, she had to **guess** whether that person had *Alzheimer's* or was *only senile*. In the past few years, however, physicians have been able to use **new and more reliable tests** focusing on genetic clues. But in the accuracy of these **new tests** lies the RISK OF ANOTHER KIND OF HUMAN TRAGEDY: physicians may be able to **predict Alzheimer's** long before its overt appearance, but such an early **diagnosis** could PSYCHOLOGICALLY DEVASTATE AN APPARENTLY HEALTHY PERSON.

That passage now coheres, or "hangs together," for three reasons:

- Its topics consistently focus on physicians and diagnosis.
- Running through it are strings of words that focus on the themes of (1) tests, (2) mental conditions, and (3) a new problem.
- And no less important, the opening sentence helps us notice those themes by emphasizing them at its end.

Again, signal a passage's key concepts in the stress positions of its opening sentences, especially the first. This principle applies to fairly long paragraphs (short, introductory, or transitional paragraphs follow different patterns). It also applies to longer passages, even whole documents.

Here's the point: We depend on concepts running through a passage to create a sense of its coherence. You help readers identify those concepts in two ways:

- Repeat those that name characters as topics of sentences, usually as subjects.
- Repeat others as themes elsewhere in a passage, in nouns, verbs, and adjectives.

Readers are more likely to notice those themes if you emphasize them at the end of the sentence that introduces the passage.

QUICK TIP For a paragraph more than five or six sentences long, underline the sentence that you think best introduces or frames the rest of the paragraph. If you can't do that quickly, your paragraph probably has a problem. If you can, circle the important words in the sentence. Those words should sound like a title for the paragraph. If they do not, your paragraph may confuse your readers. We will return to this matter in Lesson 8.

IN YOUR OWN WORDS

Exercise 6.3

Read a page of your own writing aloud, raising your voice and tapping your fingers at the ends of your sentences (as suggested in the Quick Tip on page 86). What do you notice? How often do you seem to be stressing the wrong words, and how often the right ones? Can you detect any patterns? How does your meaning change when you inadvertently stress the wrong words?

Exercise 6.4

Have a reader use the Three Tactical Revisions on pages 83–84 ("trim the end," "shift peripheral ideas to the left," or "shift new information to the right") to revise at least 4–5 sentences of your writing.

- In trimming the ends of sentences, did your reader cut material that you thought was necessary?
- Did your reader treat as peripheral ideas that you thought were important?
- Are you surprised by what your reader classified as "new information"?

Which revisions improved your writing, and which did not? Why?

SUMMING UP

Like subject and verb positions, the topic and stress positions of a sentence are relatively fixed. And just as readers want characters to align with subjects and actions to align with verbs,

so they expect to find old, familiar information at the beginning of a sentence along with its topic, and new, unfamiliar information toward the end, where it receives stress or emphasis. We can graphically integrate the principles from this lesson with our others, adding a fourth level to our sentence diagram. I call this fourth level the *attention level* because it concerns not just how easily readers understand a sentence but what they *focus on* or *attend to* in it.

Fixed Positions	Topic	Stress	Attention Level
Movable Elements	Short, simple, old	Long, complex, new	Information Level
Fixed Positions	Subject	Verb	Grammar Level
Movable Elements	Character	Action	Story Level

1. Use the end of a sentence to introduce long, complex, or otherwise difficult-to-process material, particularly unfamiliar technical terms and new information.

A determination of involvement of lipid-linked saccharides in the assembly of oligosaccharide chains of ovalbumin *in vivo* was the principal aim of this study. *In vitro* and *in vivo* studies utilizing oviduct membrane preparations and oviduct slices and the antibiotic tunicamycin were undertaken to accomplish this.

- ✓ The principal aim of this study was to determine how **lipid-linked saccharides are involved in the assembly of oligosaccharide chains of ovalbumin *in vivo***. To accomplish this, studies were undertaken *in vitro* and *in vivo*, **utilizing the antibiotic tunicamycin on preparations of oviduct membrane and on oviduct slices.**

2. Use the stress position at the very end to emphasize words that you want your readers to hear emphasized in their mind's ear:

The universities were going to have to act like businesses pursuing customers **to attract prospective students.**

- ✓ The universities were going to have to pursue students much as **businesses pursue customers.**

—Mark Edmundson

3. Use the stress of a sentence that introduces a passage to announce the key themes that the rest of the passage will develop:

In recent years, researchers have made great strides in the early and accurate **diagnosis** of *Alzheimer's disease*, but those **diagnoses** have raised a NEW PROBLEM about INFORMING THOSE MOST AT RISK WHO SHOW NO SYMPTOMS OF IT. Not too long ago, when a physician examined an older patient who seemed *out of touch with reality*, she had to **guess** whether that person had *Alzheimer's* or was *only senile*. In the past few years, however, physicians have been able to use **new and more reliable tests** focusing on genetic clues. But in the accuracy of these **new tests** lies the RISK OF ANOTHER KIND OF HUMAN TRAGEDY: physicians may be able to **predict** *Alzheimer's* long before its overt appearance, but such an early **diagnosis** could PSYCHOLOGICALLY DEVASTATE AN APPARENTLY HEALTHY PERSON.

PART THREE

Clarity of Form

Well begun is half done.

—ANONYMOUS

The beginning is half of the whole.

—PLATO